EXPERIMENT 3: INDIVIDUAL VARIATIONS IN THE RESPONSE TO DRUGS

40 STANDARD IN-COURSE MCQs

1. The major purpose of this experiment is to demonstrate:

A. Drug incompatibility B. Individual variation in drug response C. Drug excretion D. Drug absorption

Answer: B

2. The experimental drug used was:

A. Diazepam B. Pentobarbitone C. Sodium hexobarbitone D. Morphine

Answer: C

3. Sodium hexobarbitone belongs to which class?

A. Opioids B. Barbiturates C. Benzodiazepines D. Antipsychotics

Answer: B

4. The major pharmacological action of barbiturates is:

A. CNS stimulation B. CNS depression C. Peripheral vasoconstriction D. Increased skeletal muscle activity

Answer: B

5. The experimental animals used were:

A. Rats B. Rabbits C. Guinea pigs D. Albino mice

Answer: D

6. The route of administration was:

A. Oral B. Intravenous C. Intraperitoneal D. Intramuscular

Answer: C

7. The concentration of the drug was:

A. 1 mg/mL B. 5 mg/mL C. 10 mg/mL D. 20 mg/mL

Answer: C

8. Which of the following was NOT one of the parameters assessed?

A. Ataxia B. Muscle relaxation C. Response to painful stimuli D. Blood glucose

Answer: D

9. Ataxia means:

A. Increased appetite B. Loss of coordination C. Increased respiration D. Increased muscle strength

Answer: B

10. The first dose administered was:

A. 10 mg/kg B. 25 mg/kg C. 50 mg/kg D. 75 mg/kg

Answer: B

11. The highest dose administered was:

A. 75 mg/kg B. 100 mg/kg C. 125 mg/kg D. 150 mg/kg

Answer: C

12. The volume of drug administered is calculated using:

A. Dose × concentration × weight B. Dose × weight ÷ concentration C. Concentration × weight ÷ dose D. Dose ÷ weight × concentration

Answer: B

13. A mouse weighs 0.020 kg and requires 50 mg/kg of a drug with concentration 10 mg/mL. The volume required is:

A. 0.01 mL B. 0.05 mL C. 0.10 mL D. 1.0 mL

Answer: C

14. At 25 mg/kg, how many mice showed ataxia?

A. 1 B. 2 C. 4 D. 5

Answer: C

15. At 25 mg/kg, muscle relaxation occurred in:

A. 1/5 B. 2/5 C. 4/5 D. 5/5

Answer: A

16. The overall frequency of response at 25 mg/kg was:

A. 25% B. 50% C. 70% D. 100%

Answer: C

17. At 50 mg/kg, the overall frequency of response was:

A. 50% B. 65% C. 80% D. 100%

Answer: D

18. At 50 mg/kg, how many animals showed ataxia?

A. 1 B. 3 C. 4 D. 5

Answer: D

19. At 75 mg/kg, how many animals showed muscle relaxation?

A. 1 B. 3 C. 4 D. 5

Answer: D

20. At 75 mg/kg, the overall response was:

A. 50% B. 65% C. 80% D. 100%

Answer: C

21. At 100 mg/kg, muscle relaxation occurred in:

A. 1/5 B. 2/5 C. 3/5 D. 5/5

Answer: C

22. At 100 mg/kg, response to painful stimuli occurred in:

A. 0/5 B. 2/5 C. 3/5 D. 5/5

Answer: B

23. At 100 mg/kg, the overall response was:

A. 25% B. 50% C. 65% D. 80%

Answer: B

24. At 125 mg/kg, how many animals showed muscle relaxation?

A. 1 B. 2 C. 3 D. 5

Answer: D

25. At 125 mg/kg, the overall response was:

A. 50% B. 60% C. 65% D. 80%

Answer: C

26. The response pattern in this experiment was:

A. Perfectly linear B. Completely absent C. Irregular/non-linear D. Constant

Answer: C

27. Which factor can contribute to individual variation in drug response?

A. Genetics B. Receptor sensitivity C. Drug metabolism D. All of the above

Answer: D

28. Differences in hepatic enzyme activity can affect:

A. Drug metabolism B. Drug colour C. Syringe size D. Animal identification

Answer: A

29. Differences in receptor sensitivity can cause differences in:

A. Pharmacological response B. Drug concentration C. Syringe volume D. Animal weight only

Answer: A

30. Idiosyncrasy refers to:

A. Predictable dose response B. An unusual or abnormal drug response C. Drug absorption D. Drug elimination

Answer: B

31. Which of the following is an example of individual variation?

A. Two animals showing different degrees of sedation after equivalent doses B. Drug being absorbed into blood C. Drug being eliminated in urine D. Drug binding to a receptor

Answer: A

32. The graph should have dose on the:

A. Y-axis B. X-axis C. Z-axis D. No axis

Answer: B

33. Percentage frequency of response should be plotted on the:

A. X-axis B. Y-axis C. Z-axis D. Horizontal axis only

Answer: B

34. There were 5 mice and 4 response parameters. The maximum number of possible positive responses was:

A. 5 B. 10 C. 15 D. 20

Answer: D

35. If 14 of 20 possible responses are positive, the frequency is:

A. 50% B. 60% C. 70% D. 80%

Answer: C

36. One major limitation of this experiment is:

A. Very large sample size B. Small number of animals per group C. Absence of any drug D. Absence of observations

Answer: B

37. Which of the following can affect drug distribution?

A. Body composition B. Receptor number C. Genetic makeup D. Environmental temperature only

Answer: A

38. Previous exposure to a drug may result in:

A. Tolerance B. Increased syringe size C. Change in drug concentration D. Change in animal species

Answer: A

39. When drug doses are expressed logarithmically, frequency distribution of responses may approximate:

A. A square curve B. A bell-shaped normal curve C. A straight vertical line D. A triangular curve

Answer: B

40. The major conclusion of this experiment is that:

A. All individuals respond identically to drugs B. Drug response is independent of dose and individual characteristics C. Individuals can differ considerably in their responses to drugs D. Only body weight determines drug response

Answer: C